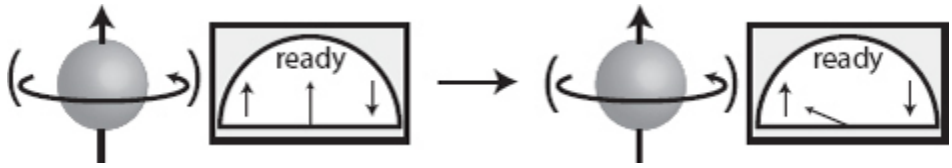
Let’s be a little more explicit about what happens when a measurement is made, according to Everett’s theory.

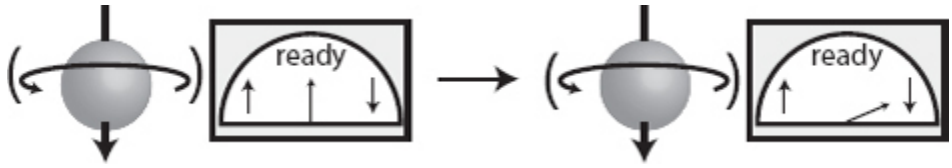
Imagine that we have a spinning electron, which could be observed to be in states of either spin-up or spin-down with respect to some chosen axis. Before measurement, the electron will typically be in some superposition of up and down. We also have a measuring apparatus, which is a quantum system in its own right. Imagine that it can be in superpositions of three different possibilities: it can have measured the spin to be up, it can have measured the spin to be down, or it might not yet have measured the spin at all, which we call the “ready” state.

The fact that the measurement apparatus does its job tells us how the quantum state of the combined spin+apparatus system evolves according to the Schrödinger equation. Namely, if we start with the apparatus in its ready state and the spin in a purely spin-up state, we are guaranteed that the apparatus evolves to a pure measured-up state, like so:



The initial state on the left can be read as “the spin is in the up state, and the apparatus is in its ready state,” while the one on the right, where the pointer indicates the up arrow, is “the spin is in the up state, and the apparatus has measured it to be up.”

Likewise, the ability to successfully measure a pure-down spin implies that the apparatus must evolve from “ready” to “measured down”:



What we want, of course, is to understand what happens when the initial spin is not in a pure up or down state, but in some superposition of both. The good news is that we already know everything we need. The rules of quantum mechanics are clear: if you know how the system evolves starting from two different states, the evolution of a superposition of both those states will just be a superposition of the two evolutions. In other words,